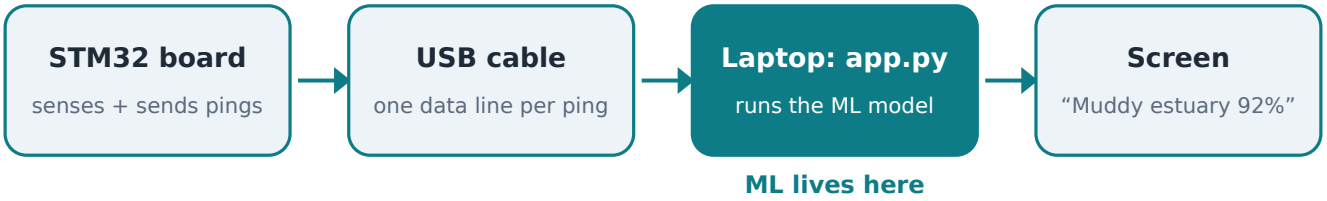
DHVANI – Where the ML Model Goes

Simple guide for the team

In one line

The ML model runs on the laptop, not on the STM32 board. The board sends data; the laptop names the water type.

How it works



What the model does

It looks at 3 inputs (temperature, salt/TDS, clarity dial) and says which of 4 water types it is:

Clear freshwater	Clear reef	Muddy river	Muddy estuary
clear + not salty	clear + salty	muddy + not salty	muddy + salty

4 steps to make it work

Step	Command	You get
1. Record data	`python ml/logger.py COM9 clear_reef 1`	Data file (do 3 rounds for each of the 4 types)
2. Train	`python ml/train.py`	dhvani_model.joblib
3. Check accuracy	`python ml/evaluate.py`	Accuracy number for the slide
4. Run dashboard	`streamlit run app.py -- COM9`	Live screen with the water-type card

Only rule: keep dhvani_model.joblib in the same folder as app.py.

Why not on the board? (if judges ask)

“The board must stay fast, simple and low-power, so it only does the ping adaptation. The ML runs at the ground station to label the water for the operator.”

Future: the model is small, so it can later be converted to C and run on the board itself.